Atharva Atul Rege

atharva-rege.github.io | github.com/Atharva-Rege | linkedin.com/in/atharva-rege-467543312/

Education

National Institute of Technology Karnataka, Surathkal — Mangaluru, India
Bachelor of Technology in ***Computer Science and Engineering – 8.94/10*** — *Aug 2023 – Present*

Private International English School — Abu Dhabi, UAE
CBSE Class 12 – ***97.6/100*** — *Aug 2022 – Mar 2023*

Experience

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) — Dec 2025 – Present
Research Intern - Mentor: Prof. Imran Razzak — *Abu Dhabi, UAE (Hybrid)*

- Introducing a X-Ray Report Generation framework using Medical World Models.
- Developed a Latent Bridge Matching framework for using non-contrast MRIs to generate contrast-enhanced MRIs in under 0.097 seconds, minimizing the time and cost required to aquire MRIs for tumor detection in MRIs.
- Collaborating with clinicians at Cleveland Clinic Abu Dhabi (CCAD) to ensure clinical relevance, dataset quality, and rigorous evaluation of results.

National Institute of Technology Karnataka, Surathkal — Feb 2026 – Present
Student Researcher - Mentor: Prof. Jeny Rajan — *Mangaluru, India (On-site)*

- Designing Mamba-based models for tumor classification in head-and-neck CT scans, focusing on robustness to label noise and limited medical data.
- Benchmarking state-of-the-art tumor classification architectures on an in-house medical imaging dataset.

I3D Lab, Indian Institute of Science (IISc) — May 2025 – July 2025
Research Intern - Mentor: Prof. Pradipta Biswas — *Bengaluru, India (On-site)*

- Implemented material property editing using CLIP embedding interpolation to modify object attributes (color, transparency, metallicity, glow, texture) in the diffusion model image generation process.
- Designed a context-aware object placement framework predicting optimal 3D position, orientation, and scale for objects within background scenes, followed by diffusion model-based inpainting.
- Achieved 97% and 99.5% mAP using YOLOv8 trained on augmented datasets generated through a two-stage synthetic data pipeline (FGVC Aircraft, Stanford Cars datasets).

Publications

- **TuLaBM: Tumor-Biased Latent Bridge Matching for Contrast-enhanced MRI Synthesis** | *MICCAI 2026 (Under Review)*
  Atharva Atul Rege, Adinath Dukre, Numan Balci, Dwarikanath Mahapatra, Imran Razzak
- **Improving MR Interaction through GenAI** | *IEEE ISMAR 2026 (Under Review)*
  Yashaswi Sinha, Yash Kumar Sahu, **Atharva Atul Rege**, Rubini Mariyappan, Subin Raj, Himanshu Vishwakarma, Mahendra Rajendra Kharsade, Abhishek Mukhopadhyay, Pradipta Biswas

Projects

The Brush of Spells : Text-Guided Image Inpainting | *Python, PyTorch* — Nov 2024 – Apr 2025

- Implemented *MMFL: Multimodal Fusion Learning for Text-Guided Image Inpainting* using Generative Adversarial Networks (GANs), leveraging a multimodal architecture to inpaint images based on textual prompts.
- Enhanced the base implementation by adding a contrastive loss and Wasserstein GAN (WGAN)-based loss to improve semantic alignment between the text and generated visuals.
- Trained the model on the CUB-200-2011 image-text dataset for better inpainting accuracy for images of birds.

Vision Kinect | *Python, OpenCV, TensorFlow, Keras* — Feb 2024 – May 2024

- Utilized YOLO object detection model to develop a hand gesture recognition system that maps each gesture to specific in-game controls, providing an interactive Tetris gaming application.
- Enabled fully touchless gameplay using gesture-based controls, demonstrating applications in interactive gaming and Human-Computer Interaction (HCI).

FinBot-AI | *Python, LangChain, Streamlit* Sep 2024 – Nov 2024

- Built a Generative AI-powered Retrieval-Augmented Generation (RAG)-based chatbot for finance-related queries, enabling accurate and context-aware responses using domain-specific knowledge sources.
- Developed an interactive web application integrated with OpenAI's GPT-4 and GPT-4o APIs to deliver real-time, user-friendly conversational financial insights.

Technical Skills

Programming Languages: Python, C, C++, Java
Machine Learning Libraries: PyTorch, TensorFlow, OpenCV, scikit-learn, NumPy, Pandas, nibabel
Frameworks: FastAPI, LangChain
Databases: MySQL, PostgreSQL
Systems: Docker, Git, Linux, Slurm
Research Interests: Deep Learning, Computer Vision, Natural Language Processing, Generative AI

Extracurricular Activities

IEEE - NITK Student Chapter Jan 2024 – Present
Executive Member *Mangaluru, India*

- Designed and coordinated visual content for social media platforms as the Computer Science Society Media Lead, enhancing engagement and strengthening the society's online presence among students and faculty.
- Mentored student-led projects titled **QuantumQuill: A Transformer-based Sci-Fi Storyteller** focused on fine-tuning Large Language Models (LLMs) to generate coherent science fiction stories, and **PathFinder: A Graph-based Career Discovery Platform** that implements a Graph Neural Network (GNN)-based temporal solution for job seekers and employers.